

Due on monday, September 26th, at the beginning of tutorial.

The questions indicated as [**SELF**] are for self-study and additional exercise. They will not be marked. So, please don't answer them in the work that you hand in. You are however strongly advised to find the time to think and work through them.

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1. (a) [**2+2 pts.**] Look up and report the definitions of
 - the hermitian conjugate of a matrix;
 - a hermitian or self-adjoint matrix.
 (Try any textbook on quantum mechanics or linear algebra, wikipedia,....)
- (b) [**4×4 pts.**] Find out whether the following matrices are hermitian or not. This will involve calculating the hermitian conjugate of each matrix and comparing with the matrix itself.

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad A = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad B = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix} \quad C = \begin{pmatrix} 0 & 1 & i \\ 1 & 1 & 0 \\ -i & 0 & 5 \end{pmatrix}$$

- (c) [**SELF**] Look up Pauli matrices. One of the above matrices is a Pauli matrix. Which one, and which Pauli matrix? What do Pauli matrices represent?

2. The equation

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) + V(x)\psi(x) = E\psi(x)$$

is the (time-independent) Schroedinger equation for a single particle in a potential $V(x)$. Solutions of this equation are stationary states of a particle in the potential $V(x)$. Note that E is a constant (independent of x) giving the energy of the stationary state.

- (a) [**2 pts.**] Write down the time-independent Schroedinger equation for the case of no potential, $V(x) = 0$.
- (b) [**6 pts.**] Show that the function $e^{-x^2/2\sigma^2}$ is not a solution of this equation for $V = 0$. You can do this by substituting $\psi(x) = e^{-x^2/2\sigma^2}$ on both sides of the equation, and then showing that the two sides cannot be equal at all x .

- (c) [8 pts.] Show that the function $\sin(kx)$ is a solution of this equation with $V = 0$, for a specific value of the energy E . Express this value of the E in terms of k , $\hbar$ and m .
- (d) [SELF] Is $\cos(kx)$ also a solution? Corresponding to the same E or different E ? What about the function e^{ikx} ?
- (e) [SELF] If $f_1(x)$ and $f_2(x)$ are both solutions, corresponding to the same value of E , then show that any linear combination $c_1 f_1(x) + c_2 f_2(x)$ is also a solution.
Does this work if $f_1(x)$ and $f_2(x)$ are both solutions, but correspond to different values E_1 and E_2 ?
- (f) [SELF] Can e^{ikx} be expressed as a linear combination of $\sin(kx)$ and $\cos(kx)$?
3. (a) [2 pts.] Now write down the time-independent Schroedinger equation for the case of a harmonic potential $V(x) = \frac{1}{2}m\omega^2 x^2$. This trapping potential is centered at $x = 0$.
- (b) [9 pts.] Show that the gaussian function $e^{-x^2/2\sigma^2}$ is a solution of this equation, for specific values of σ and E .
Indicate clearly the values of σ and E for which the gaussian function is a solution. (i.e., express σ and E as functions of $\hbar$, m and ω , such that $e^{-x^2/2\sigma^2}$ is a solution)
- (c) [3 pts.] Plot $e^{-x^2/2\sigma^2}$ as a function of x , for $\sigma = 1$ and $\sigma = 10$. Both curves should be on the same plot.
- (d) [SELF] Does σ increase or decrease with increasing strength of the trapping potential (increasing ω)? Does that make sense physically?
- (e) [SELF] Look up the expression for the energy eigenvalues (energy levels) for the quantum harmonic oscillator. (E.g., on the wikipedia page on “quantum harmonic oscillator”. Does your answer for E match one of the energy eigenvalues?)
- (f) [SELF] Now imagine that the harmonic potential is not centered at zero, but at a , i.e., $V(x) = \frac{1}{2}m\omega^2(x - a)^2$. Guess the corresponding gaussian solution to the time-independent Schroedinger equation.